Interactive data visualisation for teaching and research

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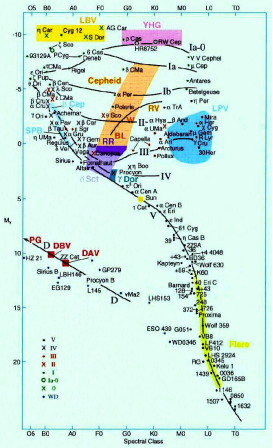
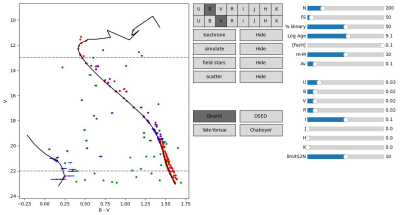
Scan this
For slides, codes, tutorials, and templates on GitHub

Why even bother?

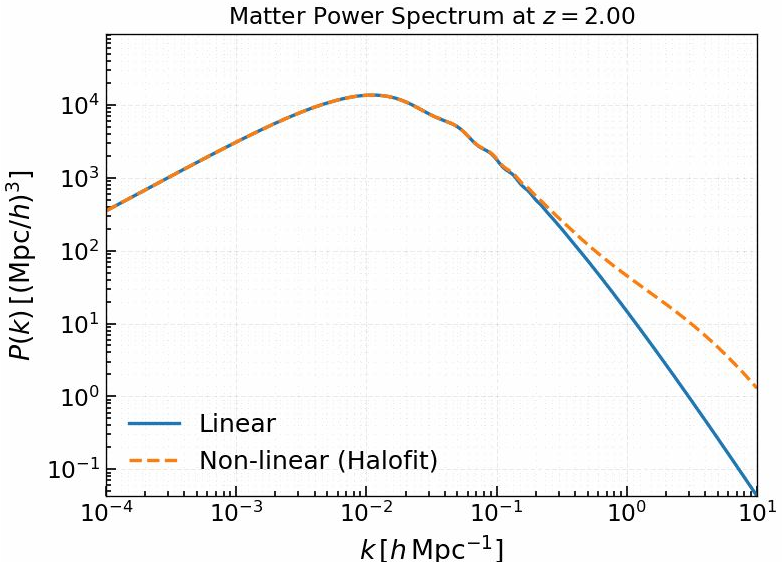
- It’s cool!
- Students learn better with interactive tools.
- Members of public feel more engaged and empowered when they can interact with the concept.
- Interactive plotting and apps are great for building intuition in your research.
- Dashboards are great for organizing your research, and help with collaborations.
- These are transferable skills even outside of academic settings.





When to use matplotlib?

- Python package for publication quality scientific plots. But also offers rudimentary interactivity.
- Good for:
  - Animations (FuncAnimation).
  - Can customize almost anything in plots.
  - Basic sliders and buttons.
- Limitations:
  - Interaction is author defined.
  - Limited native interactivity compared to web-based tools (e.g. no built-in hover/zoom UI).
  - Lots of code even for basic interaction.



Animation of Matter Powerspectrum using matplotlib.animation FuncAnimation

[Matplotlib widgets]() [Animations in Matplotlib]()

When to use Streamlit?

Streamlit

- Python package for creating beautiful web apps quickly for data exploration and communication.
- Good for:
  - Wide variety of interactivity out of the box.
  - Native support for plotting libraries like plotly, altair, etc.
  - Integration with existing python code.
  - Can host across local network, can deploy for public use on streamlit servers.
  - Easy to use (see on the right).
  - Great community support.
- Limitation:
  - Can be slow for large datasets or constant I/O operations.
- Top tip: Caching is your friend.

```
import numpy as np
import streamlit as st

def do_stuff(some_input):
      …

# streamlit supports markdown and LaTeX, and offers
different widgets to display text
st.write(“Hello, world!”)

# streamlit has many different ways to get user
input, including buttons, sliders, even uploads
my_input = st.number_input("Enter a number")

# you can use regular python scripts inside your app
x, y = do_stuff(my_input)

# streamlit also has native support for matplotlib,
plotly, altair etc.
st.line_chart(x, y)
```

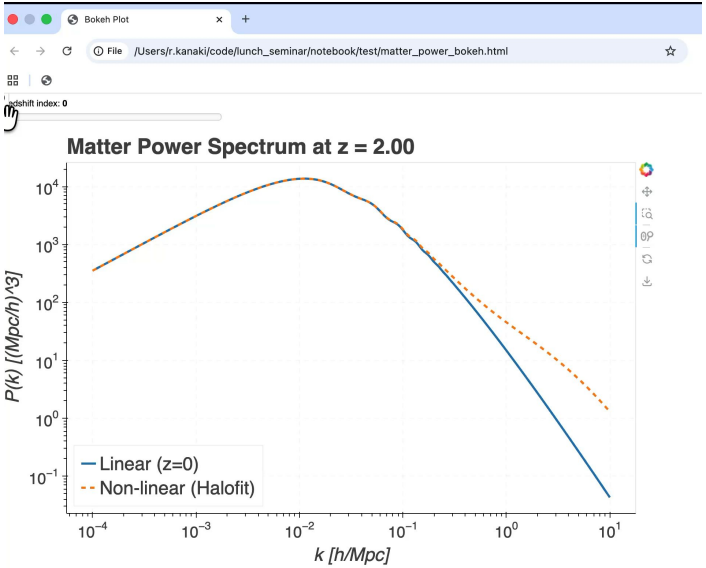
```
> streamlit run app.py
```

… in terminal will open the app in browser

Features in streamlit Streamlit extras Inspirations

When to use Plotly Dash and Bokeh?

- Python packages for creating and deploying production quality dashboards.
- Good for:
  - Fine control over complex interactions and data flows.
  - Lots of built-in interaction for plots like zoom, hover, tooltips.
  - Plotly (by itself) integrates into notebooks.
  - Well integrated with server infrastructure.
  - Bokeh is excellent for live data feeds.
- Limitations:
  - Both have steep learning curve and need you to write lots of boilerplate code.
  - Complicated to setup/host.



Interactive plot of Matter Powerspectrum using Bokeh

[Introductory notebook for Plotly](#) [Interactive app in Bokeh](#)

How to make (a robot make) an app in v0/replit ?

- LLM-based “vibe coding” platforms for developing (primarily) React apps quickly.
- Good for:
  - Rapidly prototyping an idea.
  - Creating visually stunning apps.
  - Connecting external databases (native support for free supabase upto 500MB).
  - Hosted on v0 and can do fancy things like automated scripts, traffic monitoring, etc.
- Limitations:
  - Hard to debug (or understand!).
  - Prone to hallucination.
  - 10 prompts/day (but VS code).
- Top tips: Great at understanding long contexts, good at solving their own problems.

v0 by Vercel Supabase Replit

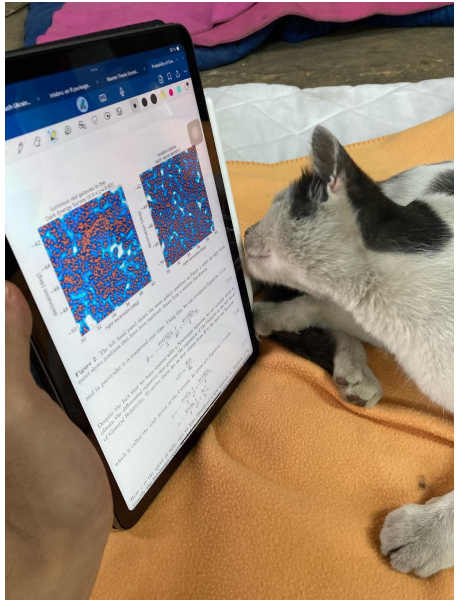
```
This is the high-level context and goal of this app.

The app needs to use following data:
A located at X, with this structure
B located at Y, with this structure and these
caveats
...
C located at Z, with this structure, but pre-process
in this way before using

The app should have these components:
 1.   Some input
 2.   A plot
 3.   ...
 4.   Another plot connected to 2 in some way

(Optionally) This is the rough outline of how to do
certain things.

The app design should be in this theme.
```

… clear instructions make better apps!

Case studies

Teaching cosmology to high school students

- **Background:**
  - Research is largely publicly funded, yet often disconnected from the general public
  - Academic environments become increasingly specialised.
- **Story:**
  - During volunteering in Kharkiv I noticed little interest in “contour plots”, but strong curiosity about cosmology (surprisingly by variety of nationality and background).
  - My own interest in physics began through public outreach.
- **Goal:** Develop a web based tool to build intuition for cosmology without advanced jargon beyond high school physics and mathematics.



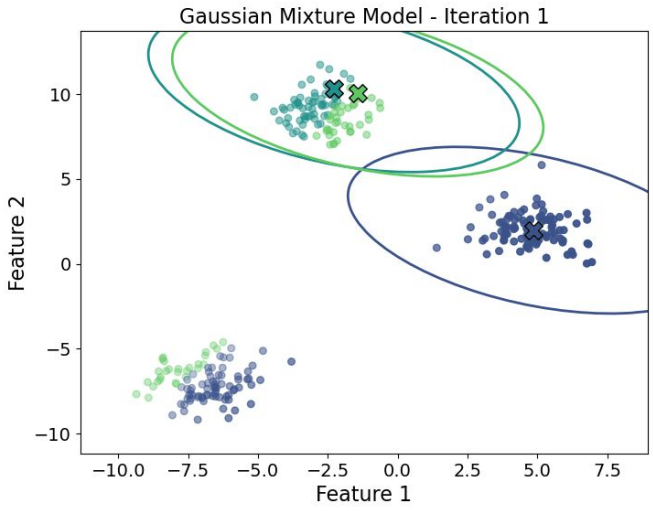
A cat in Kharkiv attempting to distinguish a random point pattern (Poisson point process) from the true galaxy distribution. So far I have succeeded in motivating a cat to study cosmology but not a dog.

Scan for the app, notes, and exercises!

Feedback is very welcome

Teaching cosmology to high school students

- **Interactive Plots & Animation**
  - Useful for building and communicating intuition, evening research contexts.
- **Examples:**
  - In industry, I worked with time-domain data(e.g. blood oxygenation). Animations helped communicate results to scientists from other backgrounds.
  - For new statistical methods, intuition is hard to gain from formulas alone, interactive visualisations makes this much easier.

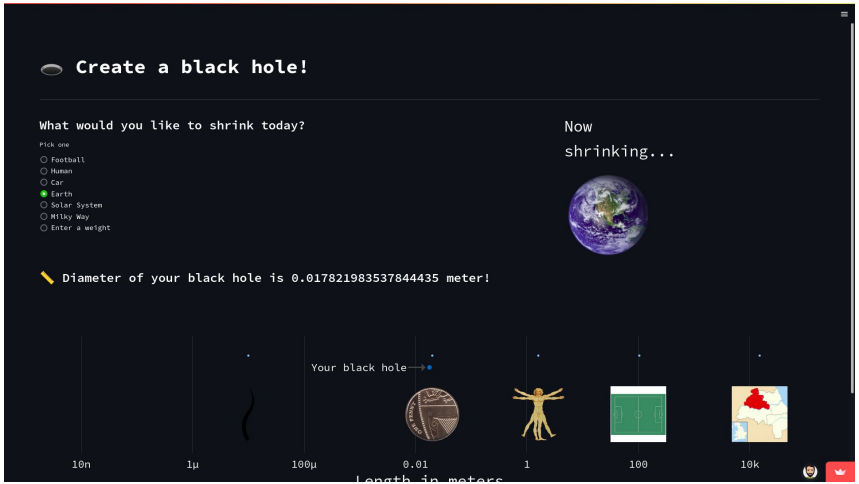


Gaussian-Mixture algorithm
Colours indicate weights (responsibility)

[Github repository for the project]

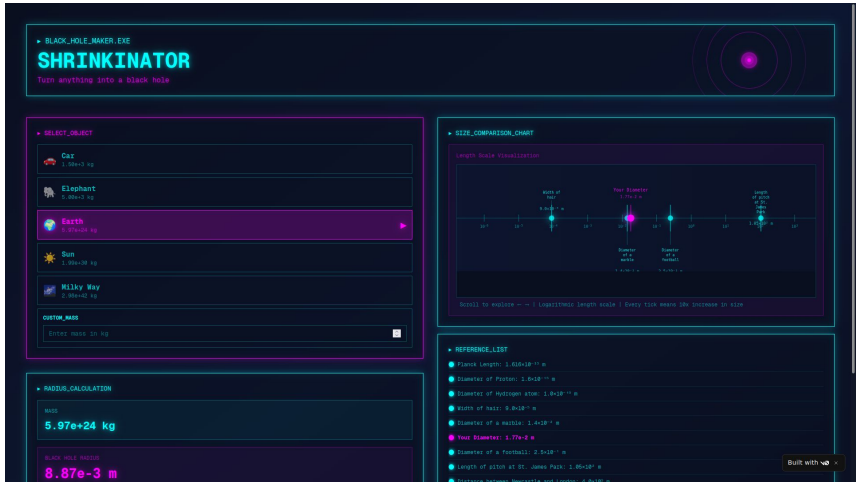
Shrinkinator - Create a black hole!

Motivation: I need some gimmick to attract people to my stall at the Great North Museum.



Made with Streamlit
(and blood and sweat): ~3 hours

What I learnt: Plotly is amazing for really fine control on interactivity of the plots



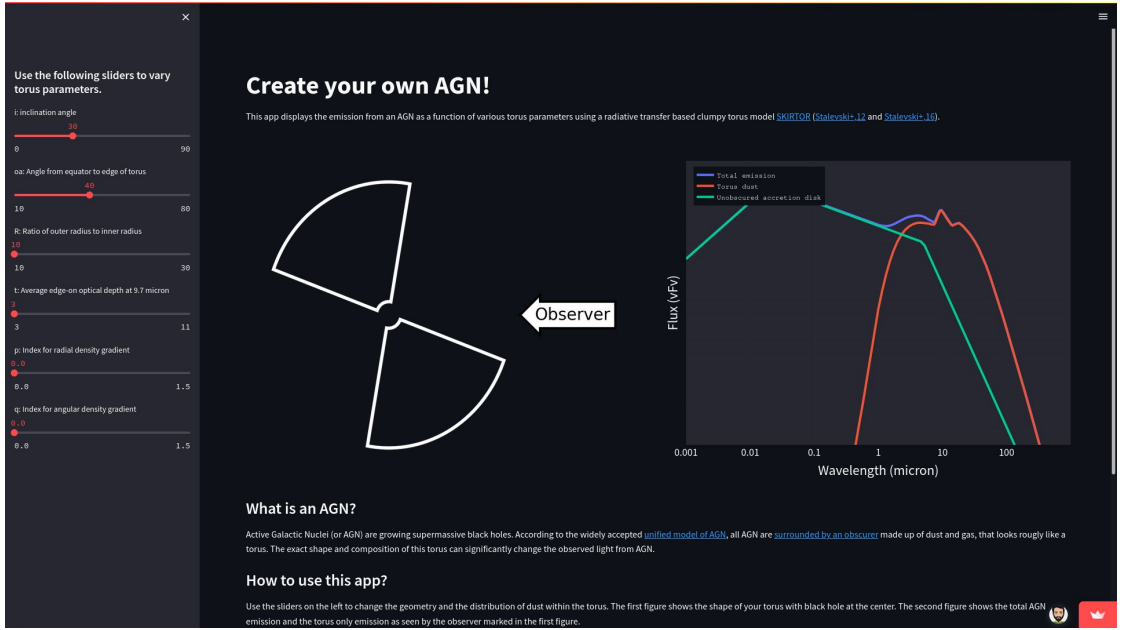
Made with robot
(and reluctance): ~20 min

What I learnt: Robots understand what “cyberpunk theme” means

Create your own Active Galactic Nucleus

Motivation: I have come across this new AGN model with lots of fancy parameters, I wonder what is the effect of each parameter on the observed SED of the AGN.

(and I need some gimmick to attract people to my poster at NAM!)



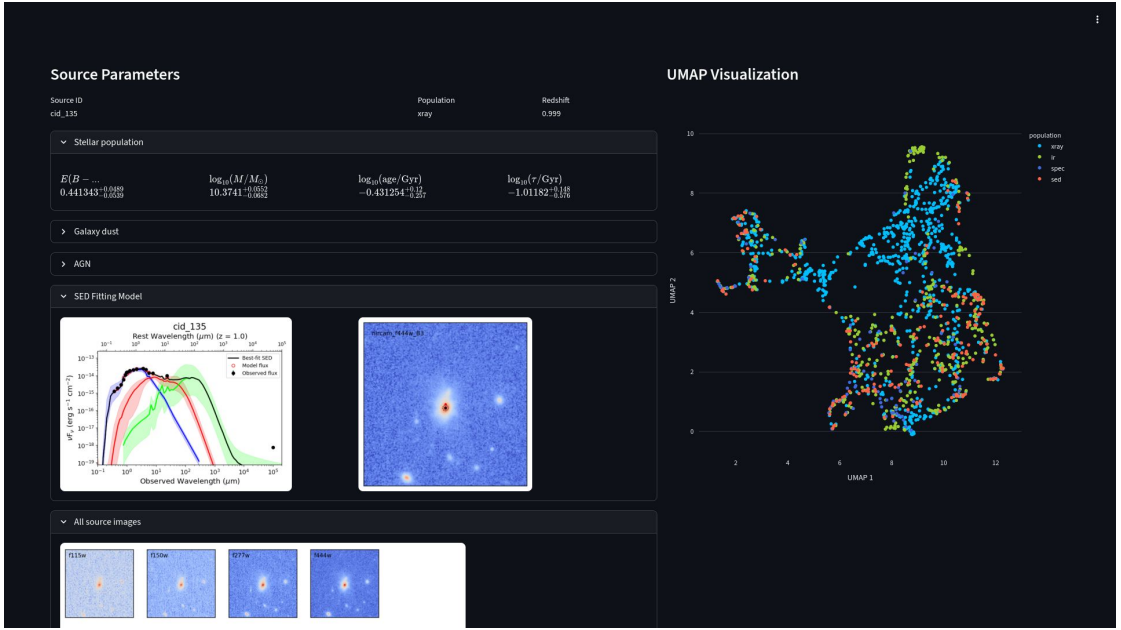
Made with Streamlit
(less blood and sweat): ~2 hours

What I learnt: You can use matplotlib and pyplot in the same app, and link them together.

Turned out to be a nice tool for explaining my research!

Improving the efficiency of research

Motivation: I have produced a complex pipeline during my PhD, but everything is cluttered everywhere. Wouldn't it be nice to have a central dashboard to quickly see everything at once?



[Made with Streamlit](#)
(and robot): ~45 min

What I learnt: Using robots (inside VS code) to do menial tasks greatly improves speed and happiness.

Streamlit supports LaTeX.

Thank you!

Create your own app!

- An app to show effect of increasing dark matter on the rotation curve of the galaxy.
- An app to show doppler shift in the spectral lines due to relative motion.
- An app to predict the type of orbit given the mass and velocity (bonus points if you also draw the orbit!).
- An app to show dip in light curve of a star due to a planet passing in front.
- … or your cool ideas! (strongly encouraged)

Scan for code snippets to get you started

https://tinyurl.com/NCLinteractive